Specialist Maths 1/2

July 2020 – Livestream Lecture

By Daniel

ABOUT ME

- Hi, I’m Daniel
- I graduated in 2017 with 99.90 ATAR
- This is a photo of me when I was doing Spesh!
- I’m currently studying Data Science at ~~Hogwarts~~ Unimelb

HOW YOU CAN BENEFIT FROM TODAY'S LECTURE

- Ask me any question via the poll section by just scrolling down in your browser

- Please get a pen & papers to brainstorm questions
- For practice questions, there'll be 60 seconds for brainstorm – so don't go away!

Overview

IN TODAY'S LECTURE

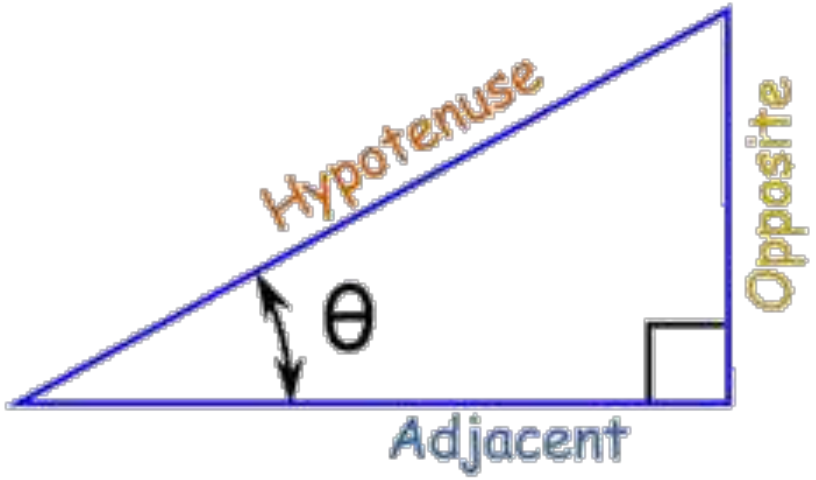
- Trigonometry
  - Break 1
- Coordinate Systems
  - Break 2
- Vectors

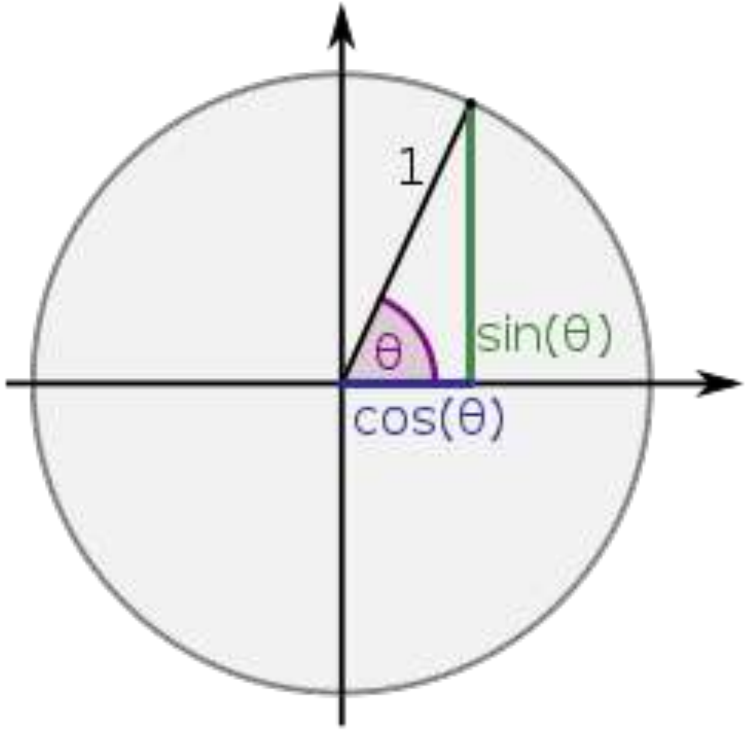
BASIC DEFINITIONS

$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

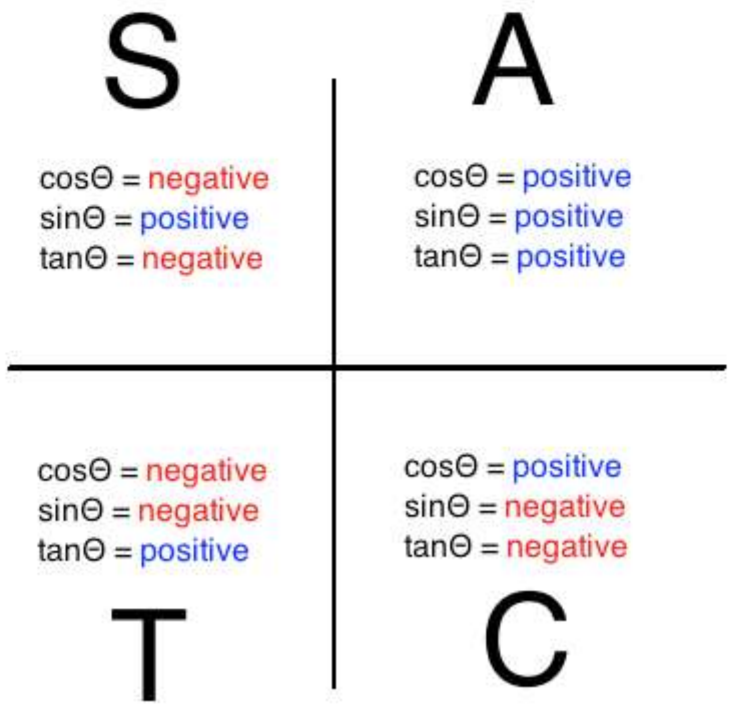
$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}}$$





A note about $\boldsymbol{\theta}$

- Positive θ are measured *anti-clockwise*
- Negative θ are measured *clockwise*
- Even though θ is typically between 0 and 360 degrees, there is strictly no limit for θ.
- $\theta > 360$ means some revolutions!



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One thing that you need to know by heart (for both Methods and Spesh) is the exact values of basic angles. Here's a trick for how to do it

	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin(x)					
cos(x)					
tan(x)					

SYMMETRICAL VALUES

- $\cos(-x) = \cos(x)$
- $\sin(\pi - x) = \sin(x)$
- $\tan(\pi + x) = \tan(x)$
- $\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$
- $\cos\left(\frac{\pi}{2} + x\right) = -\sin(x)$
- $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$
- $\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$

Evaluate $\sin(855)$

$$\sin(855)$$

$$= \sin(360\times 2 + 135)$$

$$= \sin(135)$$

$$= \sin(45 + 90)$$

$$= -\sin(45)$$

$$= -\frac{\sqrt{2}}{2}$$

Evaluate $\tan(-\frac{5\pi}{6})$

$$\tan\left(-\frac{5\pi}{6}\right)$$

$$= \tan\left(\frac{\pi}{6} - \pi\right)$$

$$= \tan\left(-\left(\pi - \frac{\pi}{6}\right)\right)$$

$$= -\tan\left(\pi - \frac{\pi}{6}\right)$$

$$= -\left(-\tan\left(\frac{\pi}{6}\right)\right)$$

$$= \tan\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{3}$$

COMPOUND & DOUBLE ANGLES

- $\sin(u \pm v) = \sin(u)\cos(v) \pm \sin(v)\cos(u)$

- $\cos(u \pm v) = \cos(u)\cos(v) \mp \sin(u)\sin(v)$

- $\sin(2x) = 2\sin(x)\cos(x)$

- $\cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1 = 1 - 2\sin^2(x)$

Evaluate $\cos(75)$

$$\cos(75)$$

$$= \cos(30+45)$$

$$= \cos(30)\cos(45) - \sin(30)\sin(45)$$

$$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{1}{2} \times \frac{\sqrt{2}}{2}$$

$$= \frac{\sqrt{6}-\sqrt{2}}{4}$$

Evaluate $\sin(5\pi/24)$

$$\cos(75) = \cos\left(\frac{5\pi}{12}\right) = \cos\left(2\times\frac{5\pi}{24}\right)$$

$$= 1 - 2\sin^2\left(\frac{5\pi}{24}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$$

$$\Rightarrow \sin^2\left(\frac{5\pi}{24}\right) = \frac{1}{2}\left(1 - \frac{\sqrt{6}-\sqrt{2}}{4}\right)$$

$$= \frac{4-\sqrt{6}+\sqrt{2}}{8}$$

$$\Rightarrow \sin\left(\frac{5\pi}{24}\right) = \sqrt{\frac{4-\sqrt{6}+\sqrt{2}}{8}} \approx 0.6088$$

Identities are equations that always hold regardless of the angle.

$$\sin^2(x) + \cos^2(x) = 1$$

$$\frac{\sin(x)}{\cos(x)} = \tan(x)$$

$$\sec^2(x) - \tan^2(x) = 1$$

$$\mathrm{cosec}^2(x) - \cot^2(x) = 1$$

$$\cot(x) = \frac{1}{\tan(x)}$$

$\sec(x) = -\frac{3}{2} \mid \frac{\pi}{2} \leq x \leq \pi$. Find $\sin(x)$

- $\sec(x) = -\frac{3}{2} \Rightarrow \cos(x) = -\frac{2}{3}$
- $\cos^2(x) + \sin^2(x) = 1 \Rightarrow \left(-\frac{2}{3}\right)^2 + \sin^2(x) = 1$
- $\Rightarrow \sin(x) = \pm\frac{\sqrt{5}}{3}$
- $\frac{\pi}{2} \leq x \leq \pi \Rightarrow \sin(x) > 0 \Rightarrow \sin(x) = \frac{\sqrt{5}}{3}$

FINAL QUESTION!

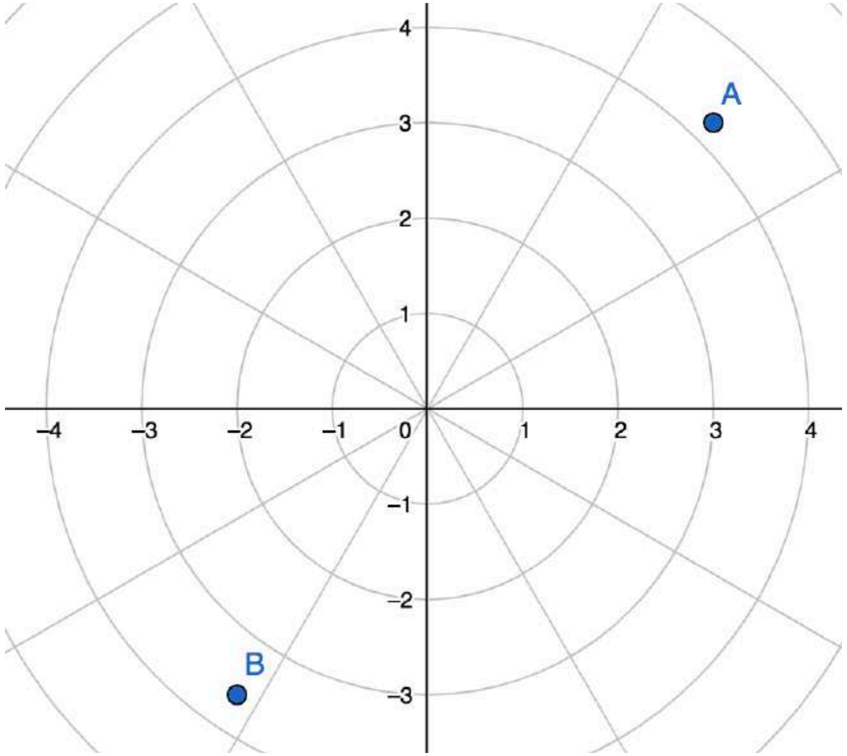
Question 5 (5 marks)

a. For the function with rule $f(x) = 96\cos(3x)\sin(3x)$, find the value of a such that $f(x) = a\sin(6x)$.

1 mark

2D

- How do we represent the position of a static point on a 2D plane ?



- How do we represent the position of a static point on a plane ?

- Using either:
  - A pair of horizontal & vertical ordinates (x, y) → Cartesian
  - A pair of distance & angle (r, θ) → Polar

- Normally, we use Cartesian system when dealing with functions
- We use Polar system when dealing with rotational quantities (vectors and complex numbers)

CONVERTING POLAR – CARTESIAN

- Polar to Cartesian:

$$x = r\cos(\theta)\ , y = r\sin(\theta)$$

  - Use trig identities if you need to!

- Cartesian to Polar:

$$r = \sqrt{x^2 + y^2}\,, \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

HARD QUESTION!

Convert the function $\boldsymbol{r = \tan(\theta)\sec(-\theta)}$ into Cartesian form

- $y = r\sin(\theta) = \tan(\theta)\sec(-\theta)\sin(\theta)$

$$= \frac{\sin(\theta)}{\cos(-\theta)} \cdot \frac{1}{\cos(\theta)} \cdot \sin(\theta) = \tan^2(\theta)$$

- $x = r\cos(\theta) = \tan(\theta)\sec(-\theta)\cos(\theta)$

$$= \frac{\sin(\theta)}{\cos(\theta)} \cdot \frac{1}{\cos(-\theta)} \cdot \cos(\theta) = \tan(\theta)$$

- $y = x^2$

IF YOU THINK CARTESIAN GRAPHING IS BAD…

https://www.desmos.com/calculator/ocpwsg5z8e

- How do we represent the path of a moving point on a plane ?
- With Cartesian system, the path is represented by a function.
- But what if we're required to find the exact position at any time t?
- Idea: instead of using (x, y) we use $\left(f(t), g(t)\right)$
- t is called the parametric variable

CONVERTING PARAMETRIC – CARTESIAN

- Parametric to Cartesian:
    - To eliminate t from both equations $x = f(t),\ y = g(t)$
    - Find $t = f^{-1}(x)$
    - Substitute into $y = g[f^{-1}(x)]$
    - Sometimes helpful to use trigonometric identities

- Cartesian to Parametric: By recognition

CONVERTING PARAMETRIC – CARTESIAN

A point moves according to the parametric equations:

$$x = t - 2$$

$$y = t^2 - 4t + 1$$

Show that the Cartesian equation of the path is $y = x^2 - 3$

$$x = t - 2 \Rightarrow t = x + 2$$

$$y = t^2 - 4t + 1$$

$$= (x+2)^2 - 4(x+2) + 1$$

$$= x^2 - 3$$

$$\Rightarrow y = x^2 - 3$$

2014 SPECMATH EXAM 1 4

Question 2 (5 marks)

The position vector of a particle at time $t \geq 0$ is given by

$$\underset{\sim}{r}(t) = (t-2)\underset{\sim}{i} + (t^2 - 4t + 1)\underset{\sim}{j}$$

a. Show that the cartesian equation of the path followed by the particle is $y = x^2 - 3$. 1 mark

Two points $A(2t - 10,3)$ and $B(2, t + 4)$ starts moving at $t = 0$.

When will their paths intersect?

- *When A intersects B:*

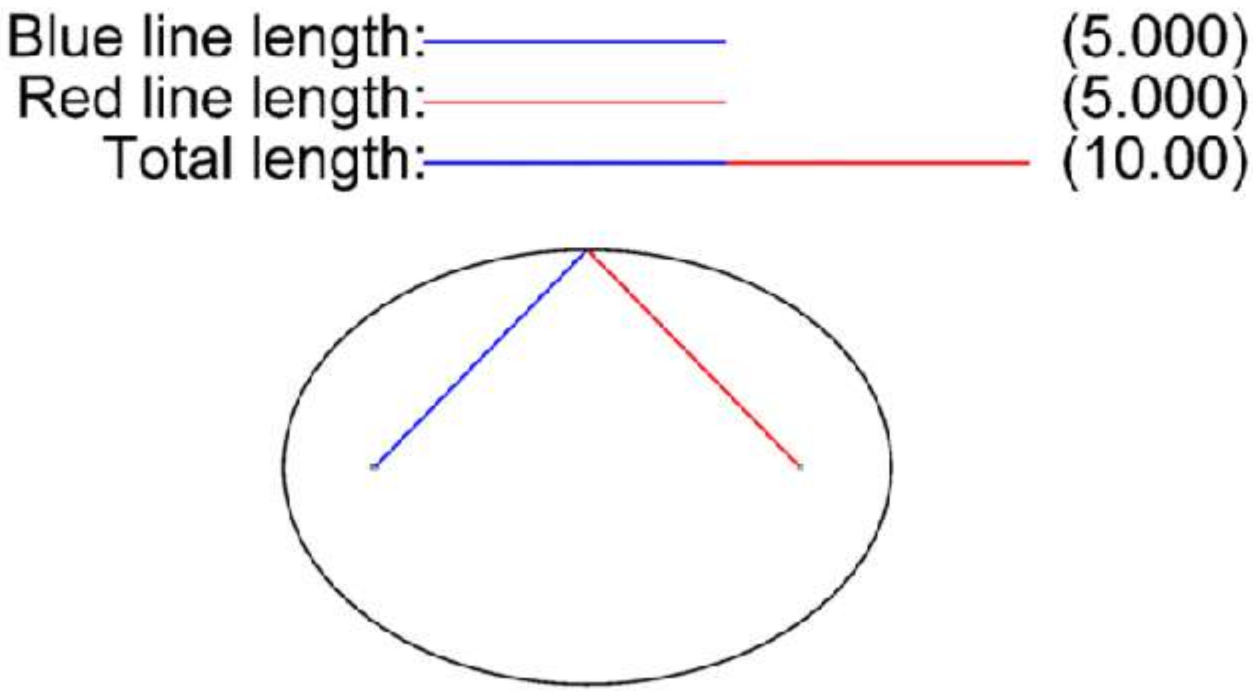
$$\begin{matrix} x_A = x_B \\ y_A = y_B \end{matrix} \Rightarrow \begin{matrix} 2t - 10 = 2 \\ 3 = t + 4 \end{matrix} \Rightarrow \begin{matrix} t = 6 \\ t = -1 \end{matrix}$$

- *There is no t such that* $(x_A, y_A) = (x_B, y_B)$*; so the paths will never intersects*

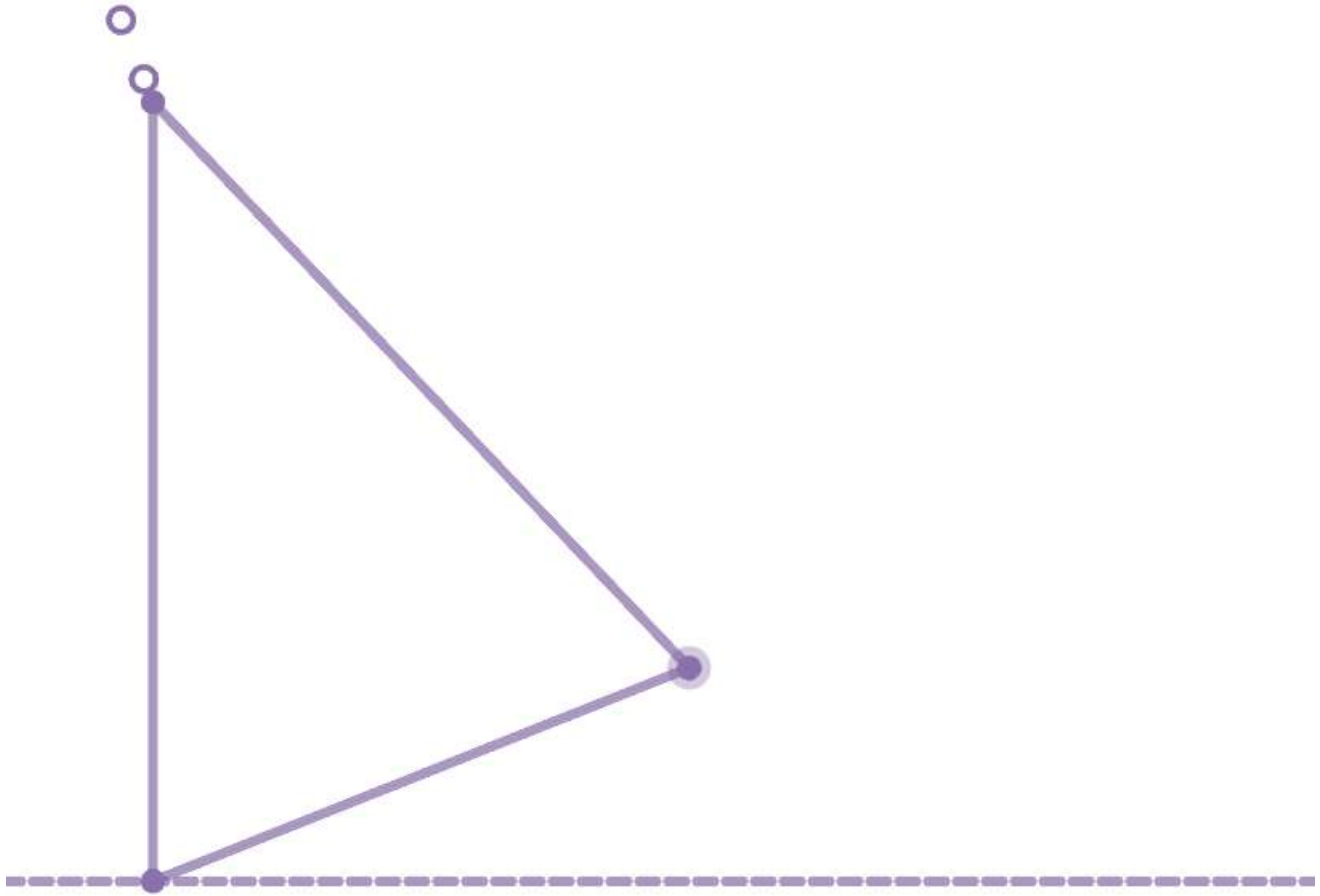
MOVING WITH A RULE

- What if the motion is defined by a specific rule?
- E.g. a point that moves in a plane so that its distance from the origin is always 3 ➔ This is a circle, centre at origin, radius 3
- The path traced by the motion of a point following a rule is called the locus of the point.
- There can be different loci for the same shape.
- Usually the rules are distance-based and regarding some fixed points

MOVING WITH A RULE



MOVING WITH A RULE



Find the Cartesian equation of the line that is equidistant from the point $\left(1, \sqrt{3}\right)$ and the origin.

Let the line be the locus of the point $P(x, y)$

Since P is equidistant from $\left(1, \sqrt{3}\right)$ and $(0,0)$:

$$\sqrt{(x-1)^2 + \left(y-\sqrt{3}\right)^2} = \sqrt{(x-0)^2 + (y-0)^2}$$

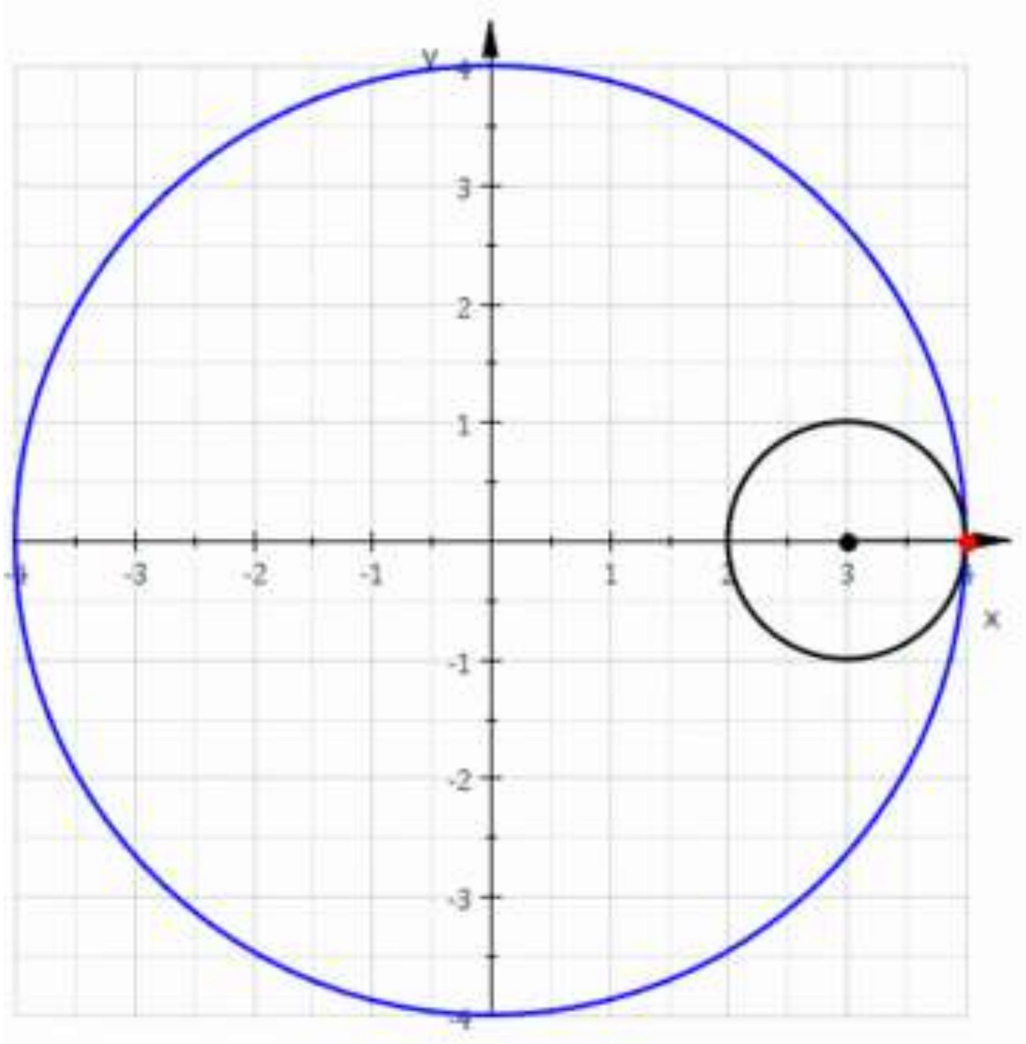
$$(x-1)^2 + \left(y-\sqrt{3}\right)^2 = x^2 + y^2$$

iii. Use the fact that the line $\left|z-\left(1+i\sqrt{3}\right)\right| = |z|$ passes through the point $z = 2$, or otherwise, to find the equation of this line in cartesian form. 1 mark

$$x^2 - 2x + 1 + y^2 - 2\sqrt{3}y + 3 = x^2 + y^2$$

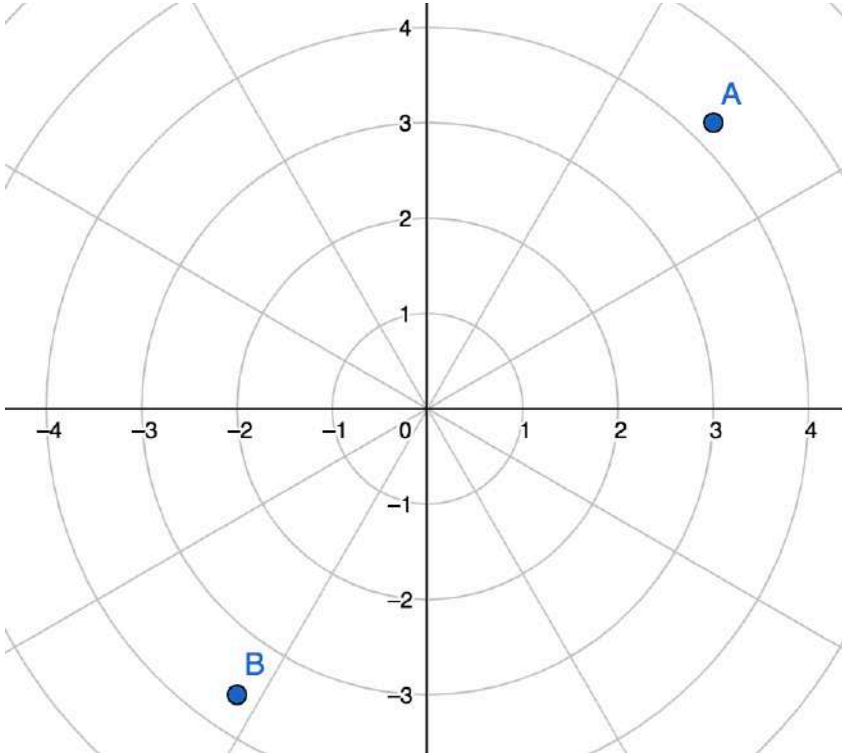
$$-2x - 2\sqrt{3}y + 3 = 0 \Rightarrow y = -\frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{2}$$

MORE MOTION!



UNPOPULAR DEFINITION

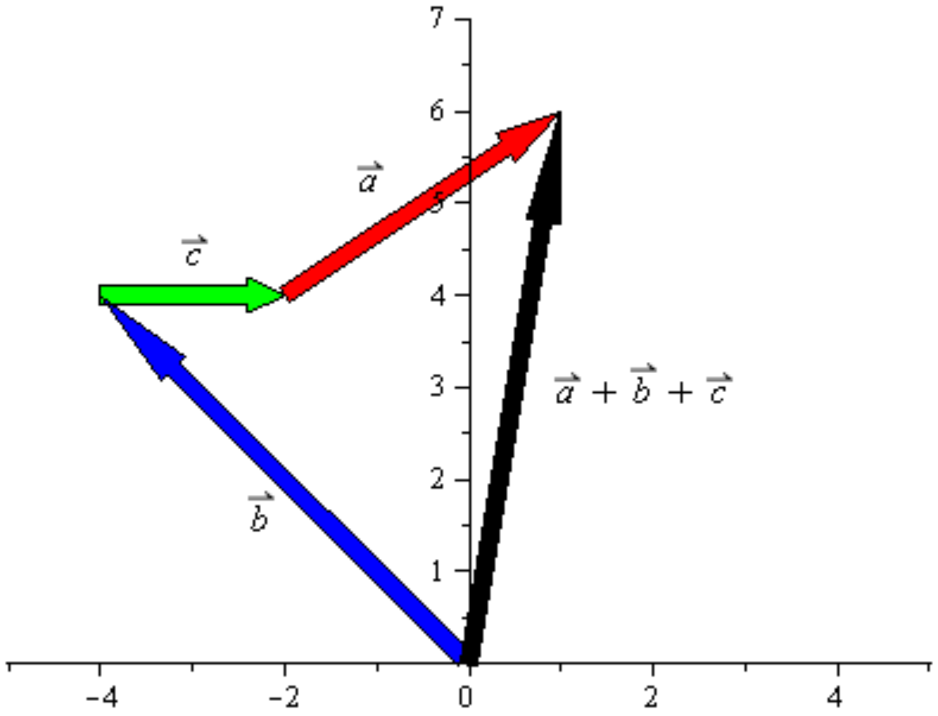
- Vectors represents the fixed straight motions of a point from origin.



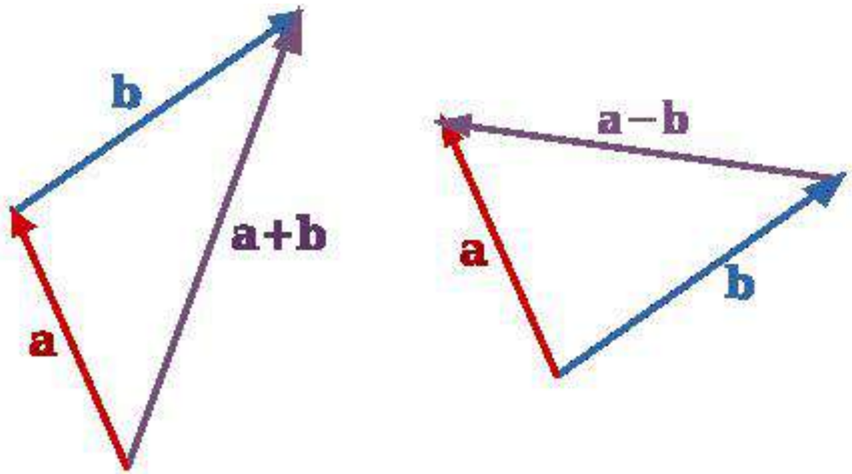
- $\vec{u} = a\vec{\imath} + b\vec{\jmath} = \begin{bmatrix} a \\ b \end{bmatrix}$

- If $A(x_A, y_A)$ and $B(x_B, y_B)$ then $\overrightarrow{AB} = (x_B - x_A)\vec{\imath} + (y_B - y_A)\vec{\jmath}$

- Magnitude (length) of a vector: $|\vec{u}| = \sqrt{a^2 + b^2}$



- $|\vec{a}| + |\vec{b}| \neq \vec{a} + \vec{b}$



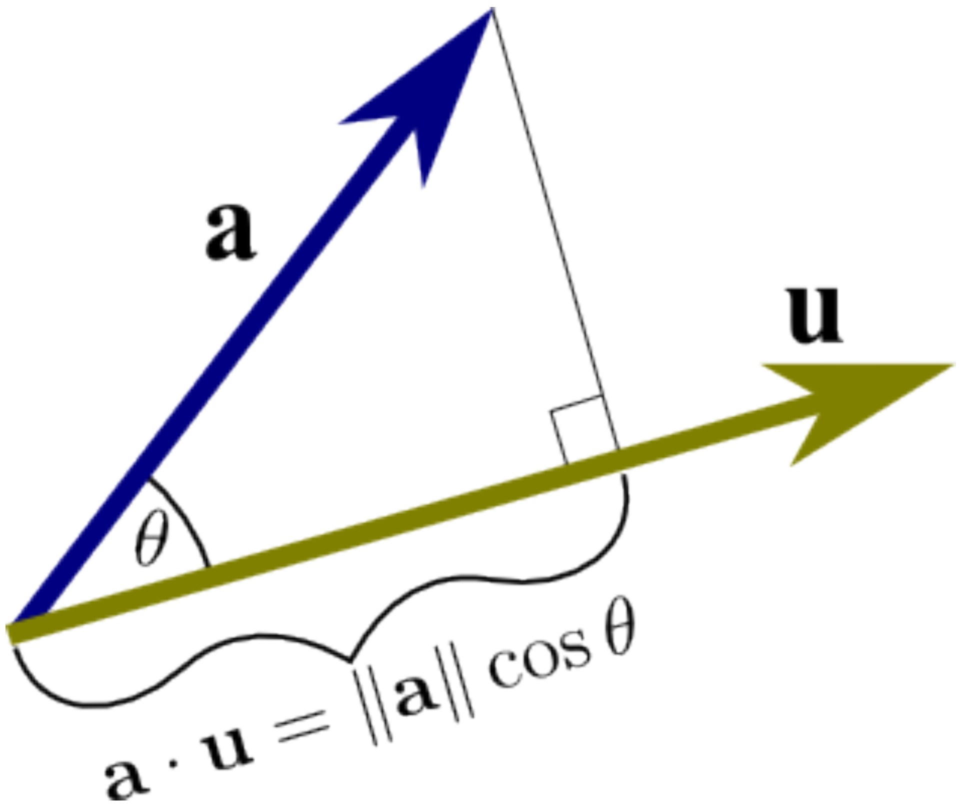
- This is a special operation for vectors. There's no geometric representation of the dot product!
- Vector "times" Vector = a Number!

$$\vec{u} \cdot \vec{v} = (a\vec{i} + b\vec{j}) \cdot (c\vec{i} + d\vec{j}) = ac + bd$$

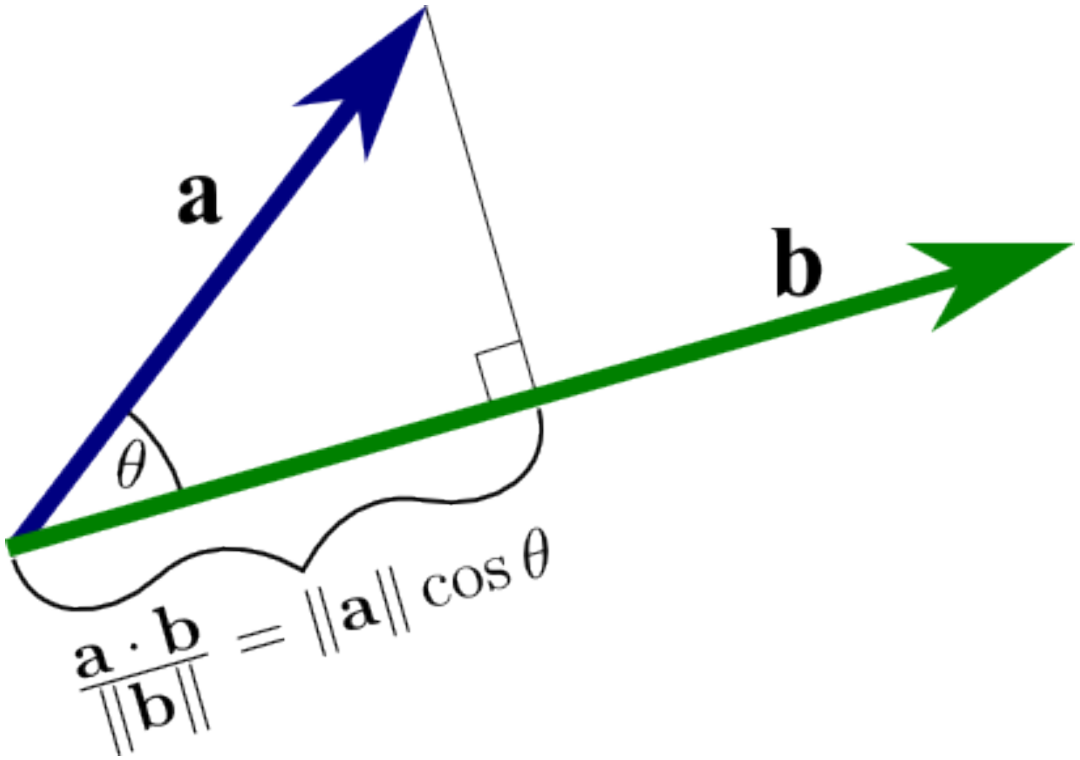
- Another formula for dot product:

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos(\vec{u}, \vec{v})$$

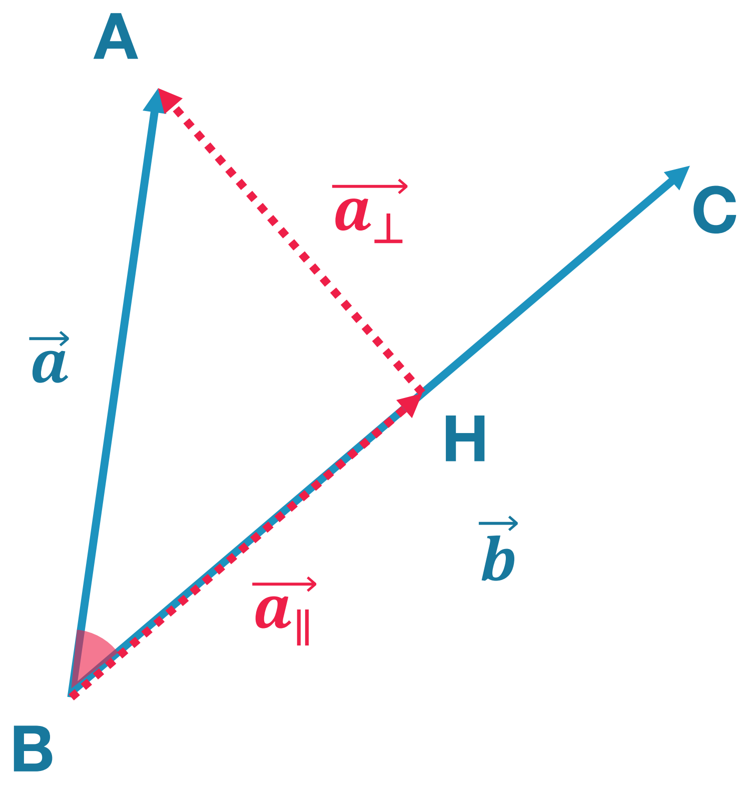
WELL, I KIND OF LIED…



- $\vec{u}$ is called the unit vector



VECTOR PROJECTION



- $\vec{a}_\perp = \vec{a} - \vec{a}_\parallel$
- $\left|\vec{a}_\parallel\right| = |\vec{a}|\cos\theta = |\vec{a}| \times \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

  $\left|\vec{a}_\parallel\right| = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

- $\vec{a}_\parallel = \left|\vec{a}_\parallel\right| \cdot \hat{b} = \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)\hat{b}$
- $\hat{b} = \frac{1}{|\vec{b}|}\vec{b}$
- $\vec{a}_\parallel = \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$

- A triangle has vertices $A\left(\sqrt{3}+1,-2,4\right)$ $B(1,-2,3)$ $C\left(2,-2,\sqrt{3}+3\right)$. Find the area of the triangle.

- $Area = \frac{1}{2} AB \cdot BC \cdot \sin(\angle ABC)$
$= \frac{1}{2}\left|\overrightarrow{AB}\right|\left|\overrightarrow{BC}\right| \cdot \sqrt{1-(\cos\angle ABC)^2}$
- $\overrightarrow{AB} = \left(1-\left(\sqrt{3}+1\right)\right)\vec{\imath} + \left(-2-(-2)\right)\vec{\jmath} + (3-4)\vec{k}$
$= -\sqrt{3}\vec{\imath} + 0\vec{\jmath} - \vec{k}$
- $\overrightarrow{BC} = \vec{\imath} + 0\vec{\jmath} + \sqrt{3}\vec{k}$

- A triangle has vertices $A\left(\sqrt{3}+1,-2,4\right)$ $B(1,-2,3)$ $C\left(2,-2,\sqrt{3}+3\right)$. Find the area of the triangle.

- $\cos\angle ABC=\dfrac{\overrightarrow{AB}\cdot\overrightarrow{BC}}{\left|\overrightarrow{AB}\right|\left|\overrightarrow{BC}\right|}=\dfrac{-2\sqrt{3}}{2\times 2}=-\dfrac{\sqrt{3}}{2}$
- $Area=\frac{1}{2}\left|\overrightarrow{AB}\right|\left|\overrightarrow{BC}\right|\cdot\sqrt{1-(\cos\angle ABC)^2}$

$=\frac{1}{2}\times 2\times 2\times\sqrt{1-\left(-\frac{\sqrt{3}}{2}\right)^2}$

$=2\times\frac{1}{2}=1$

CLOSING REMARKS

- As the school year has been somewhat disrupted, this term is going to be a bit slower so that everyone can get readjusted.
- Make sure you keep your algebra game sharp! That means solving equations, factorization and graphing.
- Focus on Vectors, Complex numbers and Kinematics / Statics – these are the ones that will be followed up in Y12.
- Have fun and enjoy :D

EMPTY SLIDES IF TIME PERMITS